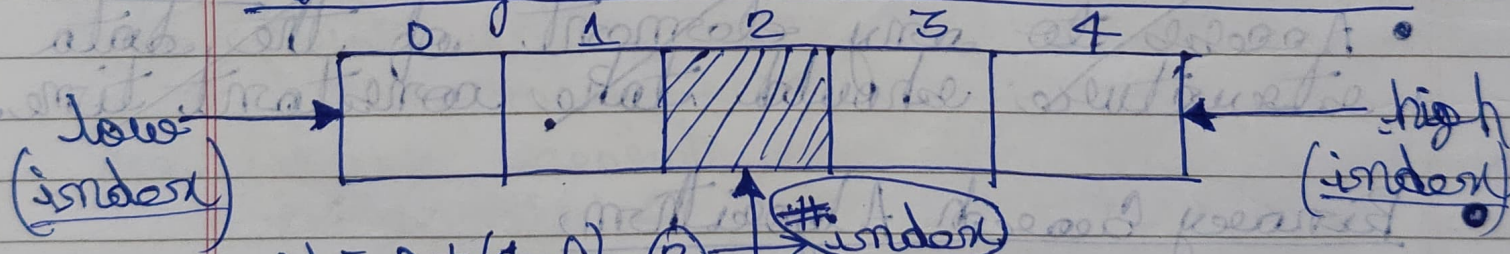


• Finding the middle index ⁶⁶ mid ⁹⁹



$$\text{mid} = 0 + \frac{(4 - 0)}{2} = 2$$

mid (index)

$$\text{mid} = \text{low} + \frac{(\text{high} - \text{low})}{2}$$

Using $\text{low} + (\text{high} - \text{low}) / 2$ instead of $(\text{low} + \text{high}) / 2$ prevents integer overflow in programming languages where integer variables have a maximum value.

Consider an array $\text{arr}[] = \{2, 5, 8, 12, 16, 23, 38, 56, 72, 91\}$, and the target = 23.

Key
23

arr =

0	1	2	3	4	5	6	7	8	9
2	5	8	12	16	23	38	56	72	91

low = 0

Mid = 4

High = 9

$$\text{mid} = \text{low} + \frac{(\text{high} - \text{low})}{2} = 0 + \frac{(9 - 0)}{2} = 4.5 = 4$$

$\Rightarrow$ Key (i.e., 23) is greater than current mid element (i.e., 16). The search space moves to the right.

Key
23

~~arr~~ arr =

0	1	2	3	4	5	6	7	8	9
2	5	8	12	16	23	38	56	72	91

low = 5

Mid = 7

High = 9

$$\text{mid} = 5 + \frac{(9 - 5)}{2} = 5 + \frac{4}{2} = 7$$

$\Rightarrow$ Key is less than the current mid 56. The search space moves to the left.

Sorted

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Key
23

	0	1	2	3	4	5	6	7	8	9		
arr	[	2	5	8	12	16	23	30	56	72	91	]

low = 5 High = 6
Mid = 5

$$\text{mid} = 5 + \frac{(6 - 5)}{2} = 5 + \frac{1}{2} = 5 + 0.5 = 5.5 = 5$$

⇒ If the Key matches the value of the mid element, the element is found and stop search.